LAB SETUP

WAZUH AGENT SOC-WIN11-LAB CONNECTED AND ACTIVE

1.AGENT-ACTIVE

```
socanalyst@NithiyaDass:/tmp$ sudo /var/ossec/bin/agent_control -l

Wazuh agent_control. List of available agents:
   ID: 000, Name: NithiyaDass (server), IP: 127.0.0.1, Active/Local
   ID: 002, Name: SOC-WIN11-LAB, IP: any, Active

List of agentless devices:

socanalyst@NithiyaDass:/tmp$
```

2. INCIDENT 01

```
socanalyst@NithiyaDass:/tmp$ sudo grep "SOC-WIN11-LAB" /var/ossec/logs/alerts/alerts.json | jq -c '{time:.timestamp, rule:.rule.id, level:.rule.level, desc:
.rule.description, mitre:.rule.mitre.id}' | grep -E '60109|60154|60111'
{"time":"2026-10-10T13:40:50.705-0400","rule":"60109","level":8,"desc":"User account enabled or created","mitre":["T1098"]}
{"time":"2026-10-10T13:40:50.721-0400","rule":"60109","level":8,"desc":"User account enabled or created","mitre":["T1098"]}
{"time":"2026-10-10T13:40:50.783-0400","rule":"60154","level":12,"desc":"Administrators Group Changed","mitre":["T1484"]}
{"time":"2026-10-10T13:40:50.828-0400","rule":"60154","level":12,"desc":"Administrators Group Changed","mitre":["T1484"]}
{"time":"2026-10-10T13:40:50.861-0400","rule":"60111","level":8,"desc":"User account disabled or deleted","mitre":["T1098","T1531"]}
{"time":"2026-10-10T13:42:45.462-0400","rule":"60109","level":8,"desc":"User account enabled or created","mitre":["T1098"]}
{"time":"2026-10-10T13:42:45.464-0400","rule":"60109","level":8,"desc":"User account enabled or created","mitre":["T1098"]}
{"time":"2026-10-10T13:42:45.473-0400","rule":"60154","level":12,"desc":"Administrators Group Changed","mitre":["T1484"]}
{"time":"2026-10-10T13:42:45.504-0400","rule":"60154","level":12,"desc":"Administrators Group Changed","mitre":["T1484"]}
{"time":"2026-10-10T13:42:45.578-0400","rule":"60111","level":8,"desc":"User account disabled or deleted","mitre":["T1098","T1531"]}
socanalyst@NithiyaDass:/tmp$
```

3.SYSMON INTEGRATION:

```
PS C:\Tools> Get-Service Sysmon64
>> Select-String -Path "C:\Program Files (x86)\ossec-agent\ossec.conf" -Pattern "Sysmon" -Context 1,2

Status   Name               DisplayName
------   ----               -----------
Running  Sysmon64           Sysmon64

IgnoreCase : True
LineNumber : 214
Line       :   <location>Microsoft-Windows-Sysmon/Operational</location>
Filename   : ossec.conf
Path       : C:\Program Files (x86)\ossec-agent\ossec.conf
Pattern    : Sysmon
Context    : Microsoft.PowerShell.Commands.MatchInfoContext
Matches    : {0}
```

4.INCIDENT 02 ENCODED POWERSHELL

```
socanalyst@NithiyaDass:/tmp$ sudo grep '"id":"92027"' /var/ossec/logs/alerts/alerts.json | tail -n 1 | jq -c '{time:.timestamp, rule:.rule.id, level:.rule.l
evel, desc:.rule.description, cmd:.data.win.eventdata.commandLine}'
{"time":"2026-10-10T14:46:49.132-0400","rule":"92027","level":4,"desc":"Powershell process spawned powershell instance","cmd":"\\\"C:\\\\WINDOWS\\\\System32
\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\\\" -enc VwByAGkAdABlAC0ASABvAHMAdAAgACIAdABlAHMAdAAiAA=="}
socanalyst@NithiyaDass:/tmp$
```

5.CUSTOM-RULE-100100

```
socanalyst@NithiyaDass:/tmp$   sudo grep '"id":"100100"' /var/ossec/logs/alerts/alerts.json | tail -n 1 | jq .
{
  "timestamp": "2026-10-10T14:49:33.938-0400",
  "rule": {
    "level": 10,
    "description": "Encoded PowerShell command executed",
    "id": "100100",
    "mitre": {
      "id": [
        "T1059.001",
        "T1027"
      ],
      "tactic": [
        "Execution",
        "Defense Evasion"
      ],
      "technique": [
        "PowerShell",
        "Obfuscated Files or Information"
      ]
    },
    "firedtimes": 1,
    "mail": false,
    "groups": [
      "sysmon",
      "powershell"
    ]
  },
  "agent": {
    "id": "002",
    "name": "SOC-WIN11-LAB",
    "ip": "192.168.154.129"
  },
  "manager": {
    "name": "NithiyaDass"
  },
  "id": "1791658173.2673782",
  "full_log": "{\"win\":{\"system\":{\"providerName\":\"Microsoft-Windows-Sysmon\",\"providerGuid\":\"{5770385f-c22a-43e0-bf4c-06f5698ffbd9}\",\"eventID\":\
"1\",\"version\":\"5\",\"level\":\"4\",\"task\":\"1\",\"opcode\":\"0\",\"keywords\":\"0x8000000000000000\",\"systemTime\":\"2026-10-10T18:49:33.2543630Z\",\
"eventRecordID\":\"667\",\"processID\":\"1424\",\"threadID\":\"4556\",\"channel\":\"Microsoft-Windows-Sysmon/Operational\",\"computer\":\"SOC-WIN11-LAB\",\"
```

```
  "id": "1791658173.2673782",
  "full_log": "{\"win\":{\"system\":{\"providerName\":\"Microsoft-Windows-Sysmon\",\"providerGuid\":\"{5770385f-c22a-43e0-bf4c-06f5698ffbd9}\",\"eventID\":\
"1\",\"version\":\"5\",\"level\":\"4\",\"task\":\"1\",\"opcode\":\"0\",\"keywords\":\"0x8000000000000000\",\"systemTime\":\"2026-10-10T18:49:33.2543630Z\",\
"eventRecordID\":\"667\",\"processID\":\"1424\",\"threadID\":\"4556\",\"channel\":\"Microsoft-Windows-Sysmon/Operational\",\"computer\":\"SOC-WIN11-LAB\",\"
severityValue\":\"INFORMATION\",\"message\":\"\\\"Process Create:\\r\\nRuleName: -\\r\\nUtcTime: 2026-10-10 18:49:33.250\\r\\nProcessGuid: {64cb6510-88bd-6a
ca-ba06-000000000b00}\\r\\nProcessId: 10032\\r\\nImage: C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\\r\\nFileVersion: 10.0.2610
0.9278 (WinBuild.160101.0800)\\r\\nDescription: Windows PowerShell\\r\\nProduct: Microsoft® Windows® Operating System\\r\\nCompany: Microsoft Corporation\\r
\\nOriginalFileName: PowerShell.EXE\\r\\nCommandLine: \\\"C:\\\\WINDOWS\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\\\" -enc VwByAGkAdABlAC0A
SABvAHMAdAAgACIAdABlAHMAdAAiAA==\\r\\nCurrentDirectory: C:\\\\Tools\\\\\\r\\nUser: SOC-WIN11-LAB\\\\socanalyst\\r\\nLogonGuid: {64cb6510-6d87-6aca-e8bb-0300
00000000}\\r\\nLogonId: 0x3BBE8\\r\\nTerminalSessionId: 1\\r\\nIntegrityLevel: High\\r\\nHashes: MD5=61F649A6FDC374A6BF36C860519E017C,SHA256=BBB6FA8C283B4D9
2120B1EF249A9B311B0F804D4CABBE9981159976C8BE76A5E,IMPHASH=2F87CE4617F593D7148B739BBF06D059\\r\\nParentProcessGuid: {64cb6510-7c93-6aca-e205-000000000b00}\\r
\\nParentProcessId: 1292\\r\\nParentImage: C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\\r\\nParentCommandLine: \\\"C:\\\\Window
s\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\\\" \\r\\nParentUser: SOC-WIN11-LAB\\\\socanalyst\\\"\"},\"eventdata\":{\"utcTime\":\"2026-10-1
0 18:49:33.250\",\"processGuid\":\"{64cb6510-88bd-6aca-ba06-000000000b00}\",\"processId\":\"10032\",\"image\":\"C:\\\\\\\\Windows\\\\\\\\System32\\\\\\\\Win
dowsPowerShell\\\\\\\\v1.0\\\\\\\\powershell.exe\",\"fileVersion\":\"10.0.26100.9278 (WinBuild.160101.0800)\",\"description\":\"Windows PowerShell\",\"produ
ct\":\"Microsoft® Windows® Operating System\",\"company\":\"Microsoft Corporation\",\"originalFileName\":\"PowerShell.EXE\",\"commandLine\":\"\\\\\\\"C:\\\\
\\\\WINDOWS\\\\\\\\System32\\\\\\\\WindowsPowerShell\\\\\\\\v1.0\\\\\\\\powershell.exe\\\\\\\" -enc VwByAGkAdABlAC0ASABvAHMAdAAgACIAdABlAHMAdAAiAA==\",\"cur
rentDirectory\":\"C:\\\\\\\\Tools\\\\\\\\\",\"user\":\"SOC-WIN11-LAB\\\\\\\\socanalyst\",\"logonGuid\":\"{64cb6510-6d87-6aca-e8bb-030000000000}\",\"logonId\
":\"0x3bbe8\",\"terminalSessionId\":\"1\",\"integrityLevel\":\"High\",\"hashes\":\"MD5=61F649A6FDC374A6BF36C860519E017C,SHA256=8BB6FA8C283B4D92120B1EF249A9B
311B0F804D4CABBE9981159976C8BE76A5E,IMPHASH=2F87CE4617F593D7148B739BBF06D059\",\"parentProcessGuid\":\"{64cb6510-7c93-6aca-e205-000000000b00}\",\"parentProc
essId\":\"1292\",\"parentImage\":\"C:\\\\\\\\Windows\\\\\\\\System32\\\\\\\\WindowsPowerShell\\\\\\\\v1.0\\\\\\\\powershell.exe\",\"parentCommandLine\":\"\\\\
\\\\\"C:\\\\\\\\Windows\\\\\\\\System32\\\\\\\\WindowsPowerShell\\\\\\\\v1.0\\\\\\\\powershell.exe\\\\\\\"\",\"parentUser\":\"SOC-WIN11-LAB\\\\\\\\socanalys
t\"}}}}",
  "decoder": {
    "name": "windows_eventchannel"
  },
  "data": {
    "win": {
      "system": {
        "providerName": "Microsoft-Windows-Sysmon",
        "providerGuid": "{5770385f-c22a-43e0-bf4c-06f5698ffbd9}",
        "eventID": "1",
        "version": "5",
        "level": "4",
        "task": "1",
        "opcode": "0",
        "keywords": "0x8000000000000000",
        "systemTime": "2026-10-10T18:49:33.2543630Z",
        "eventRecordID": "667",
        "processID": "1424",
        "threadID": "4556",
```

```
        "computer": "SOC-WIN11-LAB",
        "severityValue": "INFORMATION",
        "message": "\"Process Create:\r\nRuleName: -\r\nUtcTime: 2026-10-10 18:49:33.250\r\nProcessGuid: {64cb6510-88bd-6aca-ba06-000000000b00}\r\nProcessId
: 10032\r\nImage: C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe\r\nFileVersion: 10.0.26100.9278 (WinBuild.160101.0800)\r\nDescription: Wind
ows PowerShell\r\nProduct: Microsoft® Windows® Operating System\r\nCompany: Microsoft Corporation\r\nOriginalFileName: PowerShell.EXE\r\nCommandLine: \"C:\\
WINDOWS\\System32\\WindowsPowerShell\\v1.0\\powershell.exe\" -enc VwByAGkAdABlAC0ASABvAHMAdAAgACIAdABlAHMAdAAiAA==\r\nCurrentDirectory: C:\\Tools\\\r\nUser:
 SOC-WIN11-LAB\\socanalyst\r\nLogonGuid: {64cb6510-6d87-6aca-e8bb-030000000000}\r\nLogonId: 0x3BBE8\r\nTerminalSessionId: 1\r\nIntegrityLevel: High\r\nHashe
s: MD5=61F649A6FDC374A6BF36C860519E017C,SHA256=8BB6FA8C283B4D92120B1EF249A9B311B0F804D4CABBE9981159976C8BE76A5E,IMPHASH=2F87CE4617F593D7148B739BBF06D059\r\n
ParentProcessGuid: {64cb6510-7c93-6aca-e205-000000000b00}\r\nParentProcessId: 1292\r\nParentImage: C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershel
l.exe\r\nParentCommandLine: \"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe\" \r\nParentUser: SOC-WIN11-LAB\\socanalyst\""
      },
      "eventdata": {
        "utcTime": "2026-10-10 18:49:33.250",
        "processGuid": "{64cb6510-88bd-6aca-ba06-000000000b00}",
        "processId": "10032",
        "image": "C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe",
        "fileVersion": "10.0.26100.9278 (WinBuild.160101.0800)",
        "description": "Windows PowerShell",
        "product": "Microsoft® Windows® Operating System",
        "company": "Microsoft Corporation",
        "originalFileName": "PowerShell.EXE",
        "commandLine": "\\\"C:\\\\WINDOWS\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\\\" -enc VwByAGkAdABlAC0ASABvAHMAdAAgACIAdABlAHMAdAAiAA
==",
        "currentDirectory": "C:\\\\Tools\\\\",
        "user": "SOC-WIN11-LAB\\\\socanalyst",
        "logonGuid": "{64cb6510-6d87-6aca-e8bb-030000000000}",
        "logonId": "0x3bbe8",
        "terminalSessionId": "1",
        "integrityLevel": "High",
        "hashes": "MD5=61F649A6FDC374A6BF36C860519E017C,SHA256=8BB6FA8C283B4D92120B1EF249A9B311B0F804D4CABBE9981159976C8BE76A5E,IMPHASH=2F87CE4617F593D7148B
739BBF06D059",
        "parentProcessGuid": "{64cb6510-7c93-6aca-e205-000000000b00}",
        "parentProcessId": "1292",
        "parentImage": "C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe",
        "parentCommandLine": "\\\"C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\\\"",
        "parentUser": "SOC-WIN11-LAB\\\\socanalyst"
      }
    }
  },
  "location": "EventChannel"
}
```


6.FALSE-POSITIVE-92213

```
socanalyst@NithiyaDass:/tmp$ sudo grep '"id":"92213"' /var/ossec/logs/alerts/alerts.json | tail -n 2 | jq '{time:.timestamp, file:.data.win.eventdata.target
Filename, process:.data.win.eventdata.image, user:.data.win.eventdata.user}'
{
  "time": "2026-10-10T14:46:49.134-0400",
  "file": "C:\\\\Users\\\\socanalyst\\\\AppData\\\\Local\\\\Temp\\\\__PSScriptPolicyTest_cpczof5n.s3e.ps1",
  "process": "C:\\\\WINDOWS\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe",
  "user": "SOC-WIN11-LAB\\\\socanalyst"
}
{
  "time": "2026-10-10T14:49:33.982-0400",
  "file": "C:\\\\Users\\\\socanalyst\\\\AppData\\\\Local\\\\Temp\\\\__PSScriptPolicyTest_2dsbmdiv.2br.ps1",
  "process": "C:\\\\WINDOWS\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe",
  "user": "SOC-WIN11-LAB\\\\socanalyst"
}
```

```
socanalyst@NithiyaDass:/tmp$ sudo grep '"id":"92213"' /var/ossec/logs/alerts/alerts.json | tail -n 1 | jq '.data.win.eve
ntdata | keys'
[
  "creationUtcTime",
  "image",
  "processGuid",
  "processId",
  "targetFilename",
  "user",
  "utcTime"
]
```

7.TUNING-BEFORE-AFTER

```
socanalyst@NithiyaDass:/tmp$ sudo grep "Sysmon" /var/ossec/logs/alerts/alerts.json | tail -n 6 | jq -c '{time:.timestamp
, rule:.rule.id, level:.rule.level, desc:.rule.description, file:.data.win.eventdata.targetFilename}'
{"time":"2026-10-10T14:49:33.938-0400","rule":"100100","level":10,"desc":"Encoded PowerShell command executed","file":nu
ll}
{"time":"2026-10-10T14:49:33.982-0400","rule":"92213","level":15,"desc":"Executable file dropped in folder commonly used
 by malware","file":"C:\\\\Users\\\\socanalyst\\\\AppData\\\\Local\\\\Temp\\\\__PSScriptPolicyTest_2dsbmdiv.2br.ps1"}
{"time":"2026-10-10T14:57:41.500-0400","rule":"92058","level":12,"desc":"Application Compatibility Database launched","f
ile":null}
{"time":"2026-10-10T15:03:55.085-0400","rule":"100100","level":10,"desc":"Encoded PowerShell command executed","file":nu
ll}
{"time":"2026-10-10T15:03:55.126-0400","rule":"100101","level":3,"desc":"Tuned: PowerShell execution policy test file (b
enign, was rule 92213)","file":"C:\\\\Users\\\\socanalyst\\\\AppData\\\\Local\\\\Temp\\\\__PSScriptPolicyTest_dporsdnp.0
sl.ps1"}
{"time":"2026-10-10T15:04:12.027-0400","rule":"5402","level":3,"desc":"Successful sudo to ROOT executed.","file":null}
socanalyst@NithiyaDass:/tmp$
```

8.TROUBLESHOOTING

```
socanalyst@NithiyaDass:/tmp$ sudo /var/ossec/bin/wazuh-analysisd -t; echo "exit code: $?"
exit code: 0
socanalyst@NithiyaDass:/tmp$
```